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TDDC: A Transformer-Based Data and Knowledge Dual-Driven Scheme for Automatic Modulation Classification

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***Abstract*—Automatic Modulation Classification (AMC) plays a critical role in cognitive radio services, and latest AMC algorithms still suffers from several challenges, such as low detection accuracy, weak robustness, high dependence on labeled samples and so on in low signal-to-noise environments. To address these problems, we propose an efficient automatic modulation classification algorithm using data and knowledge dual-driven mechanism (TDDC). Initially, we design a vision transformer (ViT)-based model to extract attribute-level knowledge from the raw signal. Besides, we propose a data-driven multi-scale feature extraction model based on skip connection, to effectively capture spatio-temporal correlations of signals.Finally, to validate the effectiveness of our approach, we conduct extensive simulation experiments using the RML2016.10a and RML2016.10b datasets.**

***Index Terms*—Automatic Modulation Classification , Data-knowledge Dual-driven , Vision Transformer.**

I. Introduction

With the continuous evolution and diversification of communication technologies, AMC has emerged as a crucial component, ensuring data transmission quality and system performance [1]. The primary objective of AMC technology lies in swiftly and accurately categorize signals by scrutinizing their distinctive characteristics and patterns, thereby automatically discerning the modulation type employed in the signals. AMC plays a pivotal role across various domains, including radio spectrum monitoring, signal intelligence acquisition, smart grids, and the Internet of Things.

Existing AMC approaches are typically classified into traditional model-driven techniques and data-driven methodologies leveraging deep learning frameworks. Traditional model-driven methods encompass likelihood-based (LB) and feature-based (FB) approaches. The LB approach entails determining the modulation scheme through likelihood function optimization, renowned for its optimal performance [2], [3]. However, this method is encumbered by high computational complexity and lacks adaptability to unfamiliar channel conditions [4]. Conversely, traditional FB methods extract specific features from received signals and utilize conventional classifiers like Support Vector Machine (SVM) [5], Random Forest (RF) [6], and K Nearest Neighbors (KNN) [7]. Despite the utility of FB methods, they heavily rely on expert knowledge and struggle to model signal features in intricate dynamic transmission scenarios, resulting in suboptimal AMC performance.

In response to these challenges, a series of deep learning-based AMC algorithms have been proposed. O'Shea, Corgan, et al., pioneered the integration of convolutional neural network (CNN) models into AMC [8], marking a significant advancement in the field. In a subsequent study [9], a variant Long Short-Term Memory (LSTM) network for AMC was proposed, learning from time-domain amplitude and phase information of the modulation method in the training data without necessitating expert features like higher-order period moments. Recent research endeavors have explored innovative CNN-LSTM dual-stream architectures enhancing signal classification efficiency by leveraging I/Q channel and amplitude/phase information [10]. Liang et al. proposed CCNN-Att [11], amalgamating the attention mechanism with a complex-valued neural network to better characterize signals. Notably, recent studies have applied the ViT model [15] to AMC [12]–[14], enabling the amalgamation of global information from each sample sequence and utilizing semantically relevant data for classification. However, these data-driven AMC methods necessitate a substantial number of manually labeled samples, posing challenges for practical implementation. Moreover, prolonged model training times impede real-time performance of AMC in highly dynamic modulation-change communication scenarios.

In recent years, several research endeavors have aimed to address the limitations of purely data-driven deep learning techniques. For instance, in [16], the incorporation of industrial robot fault prior knowledge into a data-driven network was proposed to enhance fault diagnosis accuracy. Zhang et al. proposed a data-knowledge dual-driven approach of Radio Frequency Fingerprint Identification, demonstrating promising results at low SNRs [17]. Zhao et al. in [18] integrated physical properties of horizontal well permeability as prior knowledge

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TABLE I
THE SEMANTIC ATTRIBUTE KNOWLEDGE OF VARIOUS MODULATIONS.

Attribute	8PSK	BPSK	AM-SSB-SC	WBFM	GFSK
Amplitude	0	0	1	0	0
Phase	1	1	0	0	0
Frequency	0	0	0	1	1
Analog	0	0	1	1	0
Digital	1	0	0	0	1

into a data model, proposing a data-knowledge dual-driven horizontal well permeability prediction method for accurate permeability prediction in water meter wells.

To address challenges such as low signal feature modeling accuracy, dependence on labeled samples, and poor real-time detection in existing methods, we propose a novel dual-driven AMC architecture. This approach combines high-dimensional spatial attribute knowledge with visual data features to enhance AMC accuracy. Inspired by prior studies [16]–[18], we represent each modulation type's unique intrinsic attributes as binary values in Table I (1 for presence,0 for absence). For example, 8PSK is characterized by its phase and digital attributes, marked as1, while others are0. We introduce a ViT-based knowledge learning model for efficient integration of global signal information, focusing on attributes like amplitude, phase, and frequency. Additionally, we present a data-driven feature extraction model utilizing skip connections and multi-scale techniques to capture spatial correlations. Lastly, a data-knowledge embedding model is developed to transform attribute knowledge vectors and embed them in visual space. The contributions of this research can be summarized as follows:

- We design a viable data-knowledge dual-driven modeling framework, to address the challenges associated with limited labeled samples and the complexity of adapting to dynamically changing environments in practical applications;
- We propose a ViT-based knowledge extraction network, to effectively integrate the global information of learned signal sequences and achieve signal attribute knowledge;
- We construct a data-driven multi-scale feature extraction model to successfully capture feasible signal features and frequency of signals at different scales and spatial correlation of signals.

II. METHODS

A. Overall Framework

Aiming at the existing pure data-driven AMC methods in practical application scenarios with insufficient standard samples and difficulties in adapting to highly dynamic change scenarios, we propose an AMC algorithm based on a data-knowledge dual-driven architecture. As illustrated in Fig. 1, the architecture comprises three key components: a pre-trained ViT-based knowledge learning model (VKLM), a pre-trained data-Driven feature extraction model (DFEM), and a data-knowledge embedding model (DKEM). The VKLM is dedicated to the acquisition of signal attribute knowledge, while the DFEM focuses on the acquisition of features such as amplitude, phase and frequency across different scales, facilitating the learning of multi-scale representation and spatial correlation within the signals. Finally, the DKEM performs the dimensional transformation of attribute knowledge vectors and facilitates the integration of attribute knowledge with data features.

B. The ViT-based knowledge learning model

Traditional methods generally use mathematical models or machine learning methods to extract expert knowledge from signals, which generally suffer from poor feature correlation and low accuracy. To solve the above problems, we propose VKLM based on the ViT model, which mainly consists of Data Augmentation, PatchEncoder, Transformer Encoder and Global Average Pooling (GAP).

Firstly, the raw IQ signal is input to the Data Augmentation module. Since the ViT requires multiple sequences as inputs, and the raw data of the IQ signal is two sequences, the IQ signal cannot be directly input to the ViT. In this regard, we design the Data Augmentation module to achieve random scaling of the IQ signal and expand the number of channels of the original signal data to h. For the original signal X, after data augmentation, the signal sequence obtained is represented as

$$y = \left\{ \hat{X}_1, \hat{X}_2, \ldots, \hat{X}_h \right\}. \tag{1}$$

Secondly, the obtained y is input to the PatchEncoder. To satisfy the chunking requirement of patches in ViT, we convert the three-dimensional signals to two-dimensional matrices in the PatchEncoder to obtain the matrix T

$$T = \begin{bmatrix} \bar{x}_1 & \cdots & \bar{x}_{\text{Len}} \\ \check{x}_1 & \cdots & \check{x}_{\text{Len}} \\ \cdots & \cdots & \cdots \\ \bar{x}_{\text{Len}*(h-1)+1} & \cdots & \bar{x}_{\text{Len*h}} \\ \check{x}_{\text{Len}*(h-1)+1} & \cdots & \check{x}_{\text{Len*h}} \end{bmatrix}. \tag{2}$$

where matrix T is then divided into z patches $t_{z1,z2}$ of size P × Q

$$T = \begin{bmatrix} t_{1,1} & \ldots & t_{1,z_2} \\ \ldots & \ldots & \ldots \\ t_{z_1,1} & \ldots & t_{z_1,z_2} \end{bmatrix}. \tag{3}$$

Finally the patches $t_{z1,z2}$ are converted into a 1 × P*Q-sized sequence $\hat{t}_{z1,z2}$

$$\hat{t}_{z_1,z_2} = [t_{z_1,z_2}(1,:), t_{z_1,z_2}(2,:), \ldots, t_{z_1,z_2}(P,:)], \tag{4}$$

where $t_{z_1,z_2}(i,:)$ denotes row i in the patch matrix. the process diagram of Data Augmentation and PatchEncoder module is shown in Fig. 2. Since the self-attention mechanism cannot capture the positional information, it is necessary to add positional embedding before the sequence $\hat{t}_{z_1,z_2}$ after PatchEncoder to get the D-dimensional sequence $\check{t}_{z_1,z_2}$. Furthermore, the processed inputs $\check{t}_{z_1,z_2}$ are sent to the Transformer Encoder, which consists of a total of N layers of components. Each

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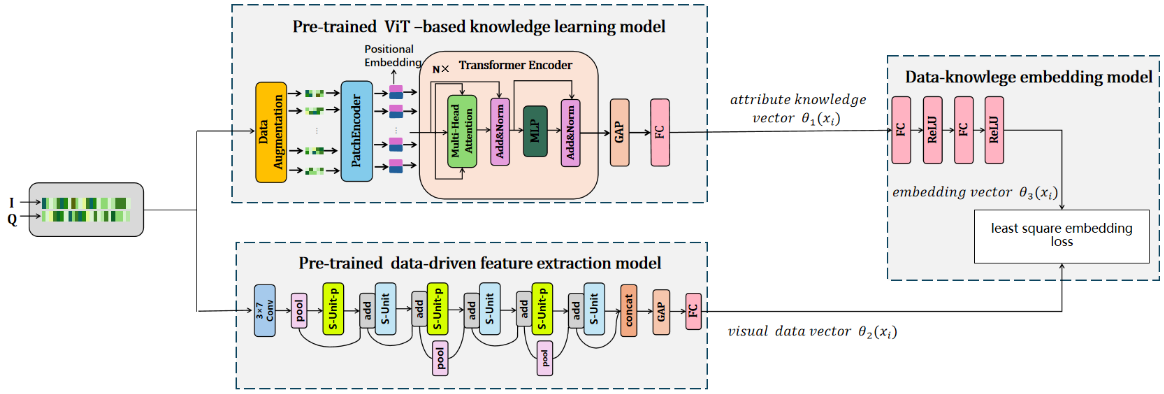

Fig. 1. The architecture of the data-knowledge dual-driven scheme

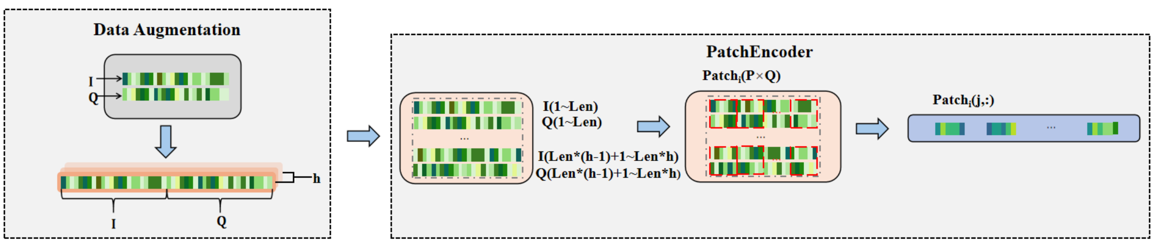

Fig. 2. The process for Data Augmentation and PatchEncoder

layer consists mainly of a Multilayer Perceptron (MLP) and a Multi-Head Attention (MHA). To maintain gradient flow and mitigate potential problems associated with gradient vanishing in deeper neural networks, a norm layer is applied after each block to preserve residual connectivity. The MHA is based on the self-attention mechanism, which computes correlations between input vectors through a scaled dot product, and generates corresponding feature vectors through a weighted summation of value vectors. The mathematical expression for this process is as follows:

$$Attention(Q, K, V) = Softmax(\frac{QK^T}{\sqrt{d_k}}) \cdot V, \tag{5}$$

where d_k stands the dimension of the key vector, and the query matrix Q, the key matrix K, and the value matrix V indicate obtained from the input sequences by means of a fully connected layer transformation, which is denoted as:

$$Q = \check{t}_{z_1,z_2} W_0 W^Q_{z_1,z_2}, K = \check{t}_{z_1,z_2} W_0 W^K_{z_1,z_2}, V = \check{t}_{z_1,z_2} W_0 W^V_{z_1,z_2}, \tag{6}$$

where W_0 represents the coefficient matrix of dimension $D \times d_{model}$.$W^Q_{z_1,z_2}$, $W^K_{z_1,z_2}$, $W^V_{z_1,z_2}$ are the coefficient matrices of dimensions $d_{model} \times d_k$, $d_{model} \times d_k$ and $d_{model} \times d_v$ respectively.

MHA is the computation of multiplying $head_i = Attention(Q_i, K_i, V_i)$ with the matrix W_0 to obtain the matrix $Head$

$$Head = [head_1, ..., head_z]W^0, \tag{7}$$

where W_0 indicates a matrix of dimension $z * d_v \times d_{model}$.

Then the output $\bar{T}$ of MHA is obtained by adding $Head$ and $\check{t}_{z_1,z_2}$. After inputting $\bar{T}$ into MLP the output of Transformer Encoder is obtained, if there are more than one layer in the Transformer Encoder then the output of the previous layer is taken as the input of the next layer. Than the output of Transformer Encoder, Global Average Pooling (GAP) is used to replace the Fully Connected (FC) layer in traditional CNN for averaging the output of different channels, with a large reduction of parameters compared to the traditional FC layer. Finally, the signal attribute knowledge vector is obtained after classification.

C. The data-driven feature extraction model

IQ signals contain a spectrum of information ranging from low to high frequencies and are characterised by complex time-frequency characteristics such as instantaneous frequency, amplitude variations and dynamic changes. In addition, IQ signals are often susceptible to noise, multipath distortion and various interferences [19], [20]. In order to improve the robustness of IQ signal recognition through a comprehensive analysis of multi-scale spatio-temporal signals, we propose a skip connection-based multi-scale feature extraction network, DFEM, which consists of two central components: S-Unit and S-Unit-p.

The S-Unit consists of four convolutional layers with different kernel sizes, shown in Fig. 3(a), arranged in parallel as 1×1, 3×1, 1×3, and 5×1, respectively. Each convolutional kernel is windowed in steps of (1,1) to ensure that the output of S-Unit preserves the full dimension of the feature map-

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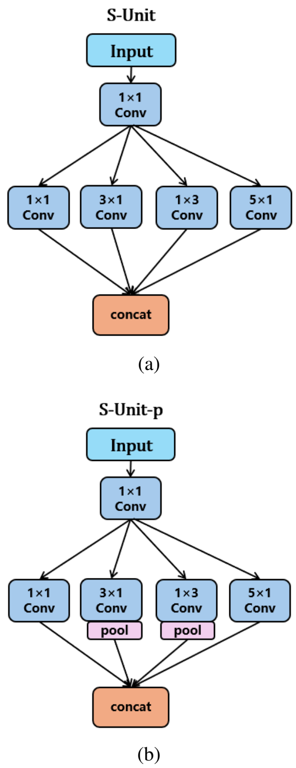

Fig. 3. (a). S-Unit. (b). S-Unit-p

ping space. In addition, S-Unit-p serves as a dimensionality reduction variant of S-Unit, integrating maximum pooling operations after the 3×1 and 1×3 convolutional layers to achieve feature map dimensionality reduction. In addition, a 1×1 convolutional layer is added on top of S-Unit-p and S-Unit to facilitate channel dimensionality reduction.

To improve the accuracy of the classification model and mitigate the negative effects of gradient vanishing induced by activation functions in deep networks, the skip connections are used to connect the S-units. Specifically, the input x of the current S-unit is connected to its output by addition. This process is formulated as follows:

$$y = \text{S-units}(x) + x, \tag{8}$$

The output from the previous unit is seamlessly propagated to the next unit via the same skip connection mechanism. In the final unit, the output y is merged with the residual information through a concat layer. After the concat operation, the FC layer of the conventional CNN is replaced by GAP to effectively aggregate multi-scale information. In particular, the GAP layer eliminates the need for parameter optimisation, thereby reducing the risk of overfitting. In this study, a signal of length 128 is used as input data. Initially, the 2×128 IQ signal in the input layer undergoes convolution with 3×7 convolutional kernels in the first layer. This first convolutional layer captures general features and facilitates further dimensionality reduction with steps of (1,2). In addition, both pooling and convolution operations are performed with steps of (1,2) at the beginning to speed up the horizontal dimensionality reduction of the feature map, thereby reducing the computational burden of subsequent layers.

D. The data-knowledge embedding model

To achieve the mutual integration of data and knowledge, we propose DKEM, which mainly consists of a transformation model and least squares embedding loss. Since the dimensionality of the two vectors obtained from VKLM and DFEM does not match, a transformation model is needed to complete the dimensionality transformation of the vectors. $\theta_1(x_i) \in \mathbb{R}^{L\times 1}$, the attribute knowledge vector obtained from VKLM, is transformed into $\theta_3(x_i) \in \mathbb{R}^{K\times 1}$, an embedding vector with the same dimensionality as that of $\theta_2(x_i) \in \mathbb{R}^{K\times 1}$, the visual data vector, after the transformation model, where the linear model has a $l2$ parameter regularisation loss for each FC layer. The last two branches are connected by a least squares embedding loss, which aims to minimise the direct difference between the visual data vector and its class representation embedding vector. The loss function is denoted as

$$L(\mathbf{W}_1, \mathbf{W}_2) = \frac{1}{N}\sum_{i=1}^{N} \|\theta_2(\mathbf{I}_i) - \theta_3(\mathbf{I}_i)\|^2 + \lambda\left(\|\mathbf{W}_1\|^2 + \|\mathbf{W}_2\|^2\right), \tag{9}$$

where N indicates the number of training samples and λ is the hyperparametric weighting of the strength of the two-parameter regularization loss with respect to the embedding loss.

III. Experiment and Analysis

A. Experimental Settings

We performed experimental evaluations on the RML2016.10a and RML2016.10b datasets. These two datasets were generated by GNU Radio [8]. RML2016.10a and RML2016.10b contain 11 and 10 modulation formats, respectively, with SNRs ranging from -20 to 18 dB in 2 dB increments.The datasets have a total of $220,000$ examples, with $1,000$ samples per SNR for each modulation format, and 128 sampling points per sample. Each sample has 128 samples. Both datasets contain actual channel defects such as channel frequency offsets, sample rate offsets, and noise to simulate imperfect transmissions. The performance of all networks in this paper is tested on a Windows 10 system equipped with a 3GHz CPU, 32GB of RAM, and an NVIDIA GeForce GTX 4070 GPU. For the two pre-trained models, we train them by adopting the SGD optimizer and train them for 80 epochs at a learning rate of 0.01. And for the data-knowledge dual-driven approach proposed in this paper, we use the Adam optimizer and Cross Entropy loss function.

B. Results and Discussions

1) Quantitative analysis: In order to validate the performance of our proposed TDDC model, we selected four SOTA models CCNN-Att [11], LSTM [9], MCNet [21] and RG

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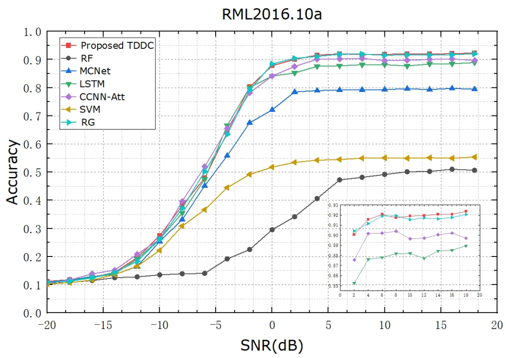

Fig. 4. Average accuracy comparisons of different SOTA methods on various SNR on the RML2016.10a.

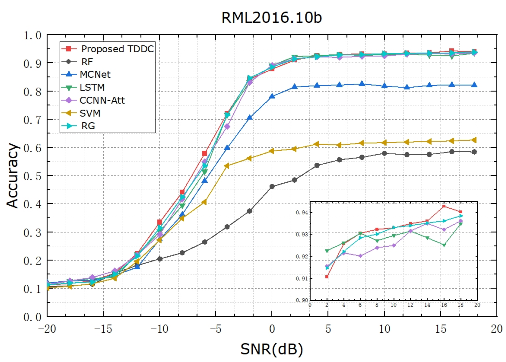

Fig. 5. Average accuracy comparisons of different SOTA methods on various SNR on the RML2016.10b.

[22] as well as two traditional AMC methods SVM [5] and RF [6] for performance comparison, and their classification performance comparison with them at each SNR is shown in Fig. 4 and Fig. 5. It can be seen that TDDC outperforms all the models in RML2016.10b when the SNR is less than 0 dB. Compared with MCNet, a well-known AMC model in recent years, our TDDC not only considers multi-scale extraction of signals, but also incorporates expert knowledge to improve the performance dramatically at low SNRs, with a gain of 10% when the SNR is equal to −4 dB. Meanwhile, our TDDC uses hopping connections to integrate the residual information during the data-driven process.

which results in a gain of 6% at a SNR of -6 dB compared to the LSTM that specializes in learning variable-length modulated signals in the time domain. And compared to two recent SOTA models, RG and CCNN-Att, our TDDC also achieves an overall incremental gain, especially at -6 dB compared to a gain of 4%. Not only that, TDDC also performs well at high SNR. The reduced sample size in RML2016.10a makes the learning efficacy of expert knowledge less effective than on RML2016.10b, resulting in a reduced gap with other models, but still outperforms all models overall. Table II shows the average accuracy of all SNRs on the two dataset experiments. It can be seen that the traditional machine learning algorithms used for AMC are far inferior to deep learning methods in

TABLE II
COMPARISONS OF AVERAGE ACCURACY ONRML2016.10A AND RML2016.10B.

Model	Accuracy in RML2016.10a	Accuracy in RML2016.10b
SVM [5]	39.53%	44.55%
RF [6]	29.60%	37.79%
CCNN-Att [11]	61.26%	63.79%
LSTM [9]	59.92%	63.71%
MCNet [21]	53.92%	56.18%
RG [22]	61.59%	64.07%
TDDC	**62.05%**	**64.56%**

processing signals in complex communication environments. And our proposed TDDC method, after combining data and knowledge, outperforms other SOTA models on both datasets, especially on the RML2016.10b dataset, and is able to get a substantial performance improvement at low SNRs, which proves the effectiveness of TDDC on AMC.

2) Qualitative analysis: The confusion matrices of TDDC for10 modulation classifications on RML2016.10b at SNRs of −6, 2, and 6 dB are presented in Fig. 6, 7, and 8, enabling a comprehensive analysis of classifier performance. In Fig. 6, confusion is notable at low SNR due to signal interference; however, AM-DSB and PAM4 are accurately recognized even at low SNR. In Figs. 7 and 8, most signals are classified accurately, with confusion primarily occurring between AM-DSB and WBFM. This confusion arises from WBFM being a higher-order modulation, which, despite facilitating high transmission rates, has closely spaced constellation points that are more susceptible to noise, complicating modulation identification.

3) Ablation Studies: To demonstrate the advantages of data-knowledge dual-driven approaches, we separately trained the pure knowledge-driven method VKLM and the pure data-driven method DFEM, and conducted recognition classification on RML2016.10b using the cross-entropy loss function. The performance comparison results are shown in Fig. 9, indicating that TDDC outperforms VKLM and DFEM not only in low SNR but also in high SNR scenarios. Overall, TDDC shows a 9% improvement over VKLM, highlighting that purely knowledge-driven approaches without data support do not yield satisfactory results, especially lagging behind TDDC by around 14% in high SNR conditions. Compared to the pure data-driven DFEM, TDDC exhibits a 6% overall improvement. The introduction of prior knowledge enhances the model's accuracy in low SNR conditions, effectively addressing the issue of low recognition accuracy due to insufficient samples.

IV. CONCLUSION

In this paper, we have proposed a data-knowledge dual-driven AMC approach. Specifically, we firstly proposed VKLM module to learn the attribute knowledge of the signal through ViT, which effectively considers the global information. Secondly, we designed a multiscale feature extraction network DFEM by using the skip connections, so as to efficiently extract the features according to the spatio-temporal

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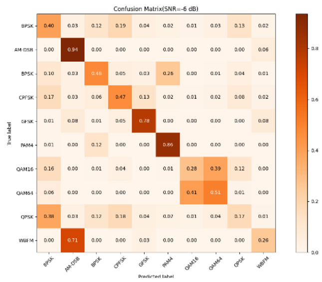

Fig. 6. Confusion matrices for the proposed method at SNR = -6 dB.

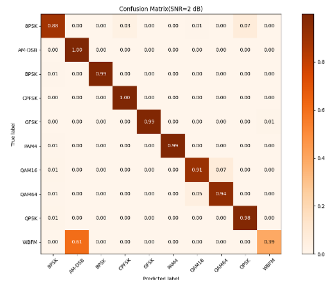

Fig. 7. Confusion matrices for the proposed method at SNR =2 dB.

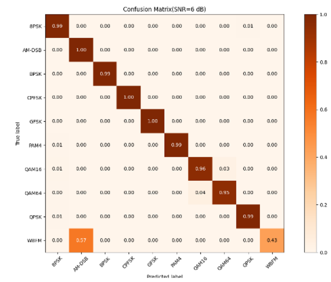

Fig. 8. Confusion matrices for the proposed method at SNR =6 dB.

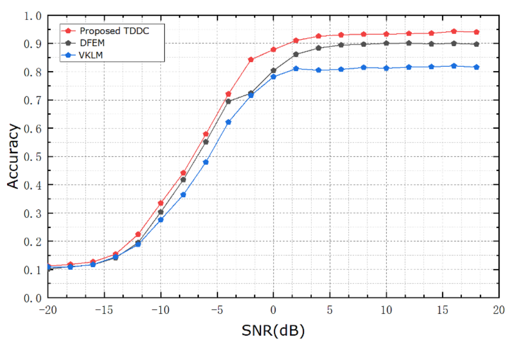

Fig. 9. Average accuracy comparisons of TDDC,VKLM and DFEM on various SNR on the RML2016.10b.

correlation of the signal. Finally, we designed DKEM module to complete the knowledge embedding of the knowledge in the visual space. The experimental results on RML2016.10a and RML2016.10b showed that the proposed method outperforms other deep learning-based AMC algorithms.

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